

# **University Campus Design & Implementation**



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BS Computer Science

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Spring 2026

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# University Network Design

(Design and Implementation Cisco Packet Tracer)

## 1. Introduction

This project presents the design and implementation of a university campus network using Cisco Packet Tracer. The objective was to build a **segmented, secure, and functional network** that supports communication among multiple departments through VLANs, inter-VLAN routing, centralized services, and access control policies.

The final network was structured around a core router, multilayer switches, access switches, end-user devices, and a dedicated data center containing service servers. The implemented services include DHCP for automatic IP addressing, DNS for name resolution, HTTP for web hosting, and FTP for file transfer between departments.

## 2. Project Objectives

- Design a complete campus network topology for a university environment.
- Implement VLAN-based segmentation for different departments.
- Enable communication using inter-VLAN routing.
- Configure centralized services including DHCP, DNS, HTTP, and FTP.
- Apply security using ACLs and port security.
- Verify network functionality through connectivity and service tests.

## 3. Network Overview

The network represents a medium-sized university environment with multiple functional areas. Separate VLANs and subnets were assigned to each department to improve organization, security, and traffic isolation.

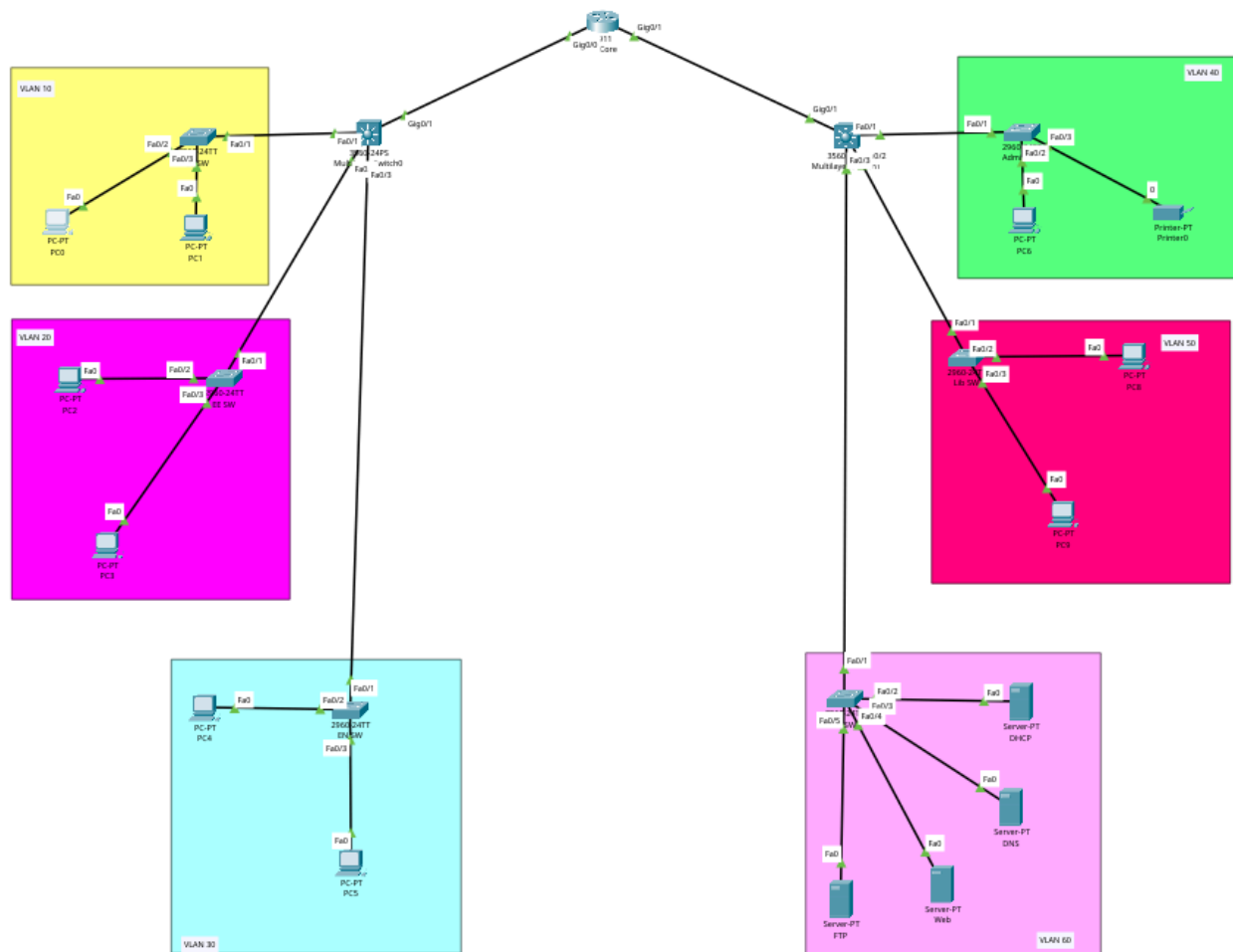
The implemented departments in the final design are Computer Science, Electrical Engineering, Engineering, Administration, Library, and Data Center. The data center hosts the servers that provide network-wide services to the clients.

## 4. Topology Design

The topology follows a hierarchical design. A Cisco 2911 router is used as the core routing device between the two multilayer distribution switches. Cisco 3560 multilayer switches are used for Layer 3 switching and VLAN gateway functions, while Cisco 2960 switches are used at the access layer for PCs, printers, and servers.

This hierarchical model improves manageability and makes the network easier to troubleshoot and document. It also reflects a realistic campus-style structure used in academic and enterprise environments.

Figure 1: High Level Architecture



# 5. Device List

The following devices were used in the implementation:

- 1 x Cisco 2911 Router (Core Router)
- 2 x Cisco 3560 Multilayer Switches (Distribution Layer)
- 6 x Cisco 2960 Switches (Access Layer)
- Multiple PCs for end-user departments
- 1 x Printer for Administration
- 4 x Servers for DHCP, DNS, Web, and FTP services

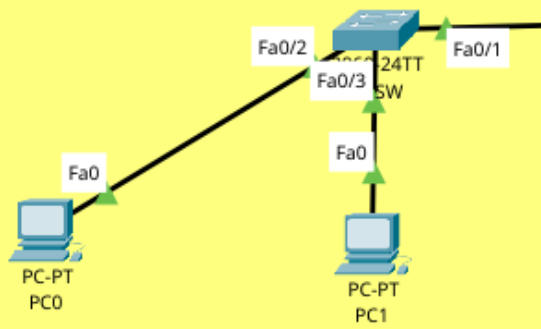
# 6. VLAN Design

Each department was assigned a dedicated VLAN to achieve traffic separation and logical grouping. The VLAN plan used in the final implementation is shown below.

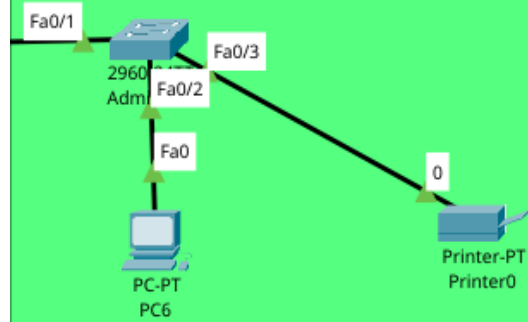
Table 1: IP Addressing Plan for VLAN's

| Department             | VLAN ID | Subnet          | Default Gateway |
|------------------------|---------|-----------------|-----------------|
| Computer Science       | 10      | 192.168.10.0/24 | 192.168.10.1    |
| Electrical Engineering | 20      | 192.168.20.0/24 | 192.168.20.1    |
| English Department     | 30      | 192.168.30.0/24 | 192.168.30.1    |
| Administration         | 40      | 192.168.40.0/24 | 192.168.40.1    |
| Library                | 50      | 192.168.50.0/24 | 192.168.50.1    |
| Data Center            | 60      | 192.168.60.0/24 | 192.168.60.1    |

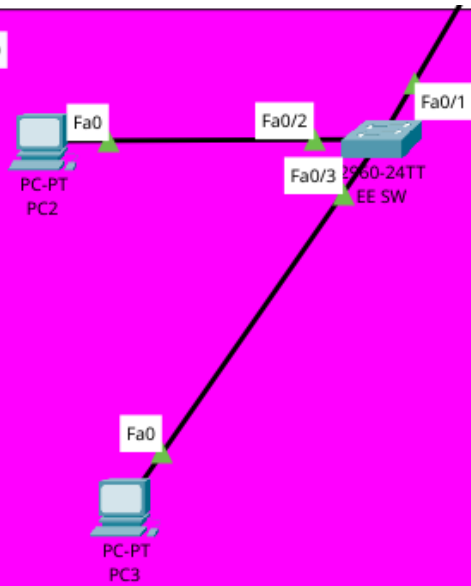
VLAN 10



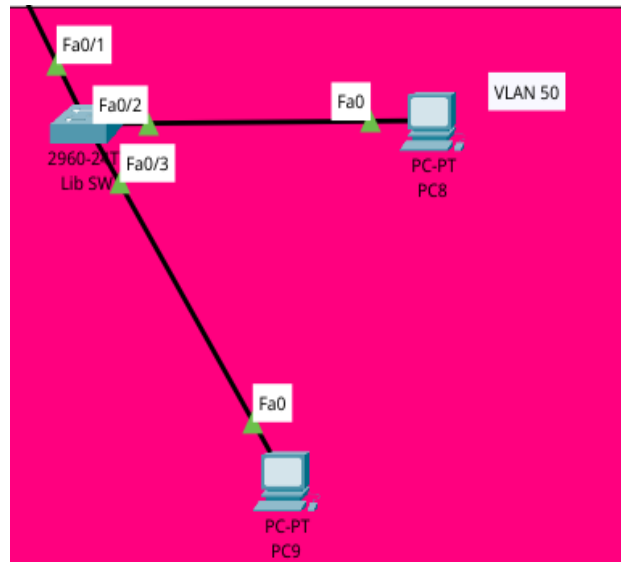
VLAN 40

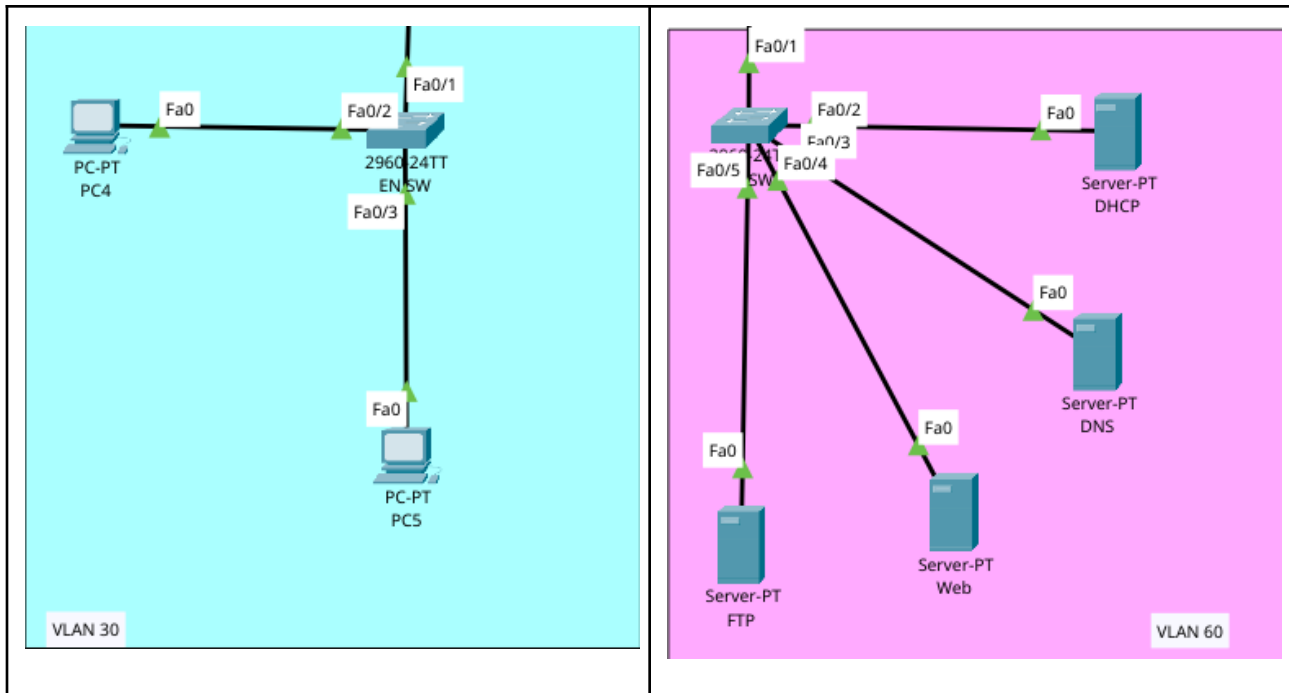


VLAN 20



VLAN 50





## 7. IP Addressing Scheme

A private IPv4 addressing scheme based on 192.168.x.0/24 networks was used. Each VLAN received a separate subnet, and the corresponding VLAN interface on the multilayer switch was configured as the default gateway.

Point-to-point links between the core router and multilayer switches were configured using /30 subnets to support routed communication between the left and right parts of the campus network.

## 8. Routing Protocol Configuration

The campus network uses Open Shortest Path First (OSPF) as the dynamic routing protocol instead of static routing. OSPF was selected because it is scalable, easier to manage in multi-network environments, and more suitable for a campus topology where multiple VLANs and routed links must exchange reachability information automatically.

In this design, a single-area OSPF configuration (Area 0) was implemented between the core router and the two multilayer switches. The router acts as the backbone device connecting both sides of the campus, while each multilayer switch advertises its locally connected VLAN networks into the OSPF domain.

The left multilayer switch advertises the Computer Science, Electrical Engineering, and Engineering VLANs, while the right multilayer switch advertises the Administration, Library, and Data Center VLANs. The point-to-point links between the router and the multilayer switches were also included in OSPF so that all Layer 3 devices could form neighbor relationships and exchange routes dynamically.

The use of OSPF eliminated the need to manually configure and maintain static routes for each remote subnet. Once OSPF adjacency was established, all remote networks were learned automatically and appeared in the routing table with the route code O, indicating OSPF-learned routes

```
Switch#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/30 is subnetted, 2 subnets
C       10.0.0.0 is directly connected, GigabitEthernet0/1
O       10.0.0.4 [110/2] via 10.0.0.2, 00:05:40, GigabitEthernet0/1
C     192.168.10.0/24 is directly connected, Vlan10
C     192.168.20.0/24 is directly connected, Vlan20
C     192.168.30.0/24 is directly connected, Vlan30
O     192.168.40.0/24 [110/3] via 10.0.0.2, 00:04:52, GigabitEthernet0/1
O     192.168.50.0/24 [110/3] via 10.0.0.2, 00:04:41, GigabitEthernet0/1
O     192.168.60.0/24 [110/3] via 10.0.0.2, 00:04:41, GigabitEthernet0/1

Switch#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
10.0.0.6       1     FULL/DR         00:00:31    10.0.0.2       GigabitEthernet0/1
Switch#
```

## Multi-Layer Switch 1



```
Switch#show ip ospf neighbor
```

| Neighbor ID | Pri | State   | Dead Time | Address  | Interface          |
|-------------|-----|---------|-----------|----------|--------------------|
| 10.0.0.6    | 1   | FULL/DR | 00:00:30  | 10.0.0.6 | GigabitEthernet0/1 |

```
Switch#show ip route
```

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
\* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/30 is subnetted, 2 subnets
O    10.0.0.0 [110/2] via 10.0.0.6, 00:00:32, GigabitEthernet0/1
C    10.0.0.4 is directly connected, GigabitEthernet0/1
S    192.168.10.0/24 [1/0] via 10.0.0.6
S    192.168.20.0/24 [1/0] via 10.0.0.6
S    192.168.30.0/24 [1/0] via 10.0.0.6
C    192.168.40.0/24 is directly connected, Vlan40
C    192.168.50.0/24 is directly connected, Vlan50
C    192.168.60.0/24 is directly connected, Vlan60
```

## Multi-Layer Switch 2

```
Router#show ip ospf neighbor
```

| Neighbor ID  | Pri | State    | Dead Time | Address  | Interface          |
|--------------|-----|----------|-----------|----------|--------------------|
| 192.168.30.1 | 1   | FULL/BDR | 00:00:35  | 10.0.0.1 | GigabitEthernet0/0 |
| 192.168.60.1 | 1   | FULL/BDR | 00:00:30  | 10.0.0.5 | GigabitEthernet0/1 |

```
Router#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
\* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet0/0
L    10.0.0.2/32 is directly connected, GigabitEthernet0/0
C    10.0.0.4/30 is directly connected, GigabitEthernet0/1
L    10.0.0.6/32 is directly connected, GigabitEthernet0/1
O    192.168.10.0/24 [110/2] via 10.0.0.1, 00:04:32, GigabitEthernet0/0
O    192.168.20.0/24 [110/2] via 10.0.0.1, 00:04:32, GigabitEthernet0/0
O    192.168.30.0/24 [110/2] via 10.0.0.1, 00:04:32, GigabitEthernet0/0
O    192.168.40.0/24 [110/2] via 10.0.0.5, 00:03:58, GigabitEthernet0/1
O    192.168.50.0/24 [110/2] via 10.0.0.5, 00:03:48, GigabitEthernet0/1
O    192.168.60.0/24 [110/2] via 10.0.0.5, 00:03:48, GigabitEthernet0/1
```

```
Router#
```

## Router

## 9. DHCP Configuration

A centralized DHCP server was placed in the Data Center VLAN. Separate DHCP pools were created for each client VLAN so that end devices could automatically receive IP addresses, subnet masks, default gateways, and the DNS server address.

Because the DHCP server is located in a different VLAN, ip helper-address was configured on the VLAN interfaces of the multilayer switches to relay DHCP requests from clients to the server.

| Pool Name  | Default Gateway | DNS Server   | Start IP Address | Subnet Mask   | Max User | TFTP Server | WLC Address |
|------------|-----------------|--------------|------------------|---------------|----------|-------------|-------------|
| LIB        | 192.168.50.1    | 192.168.60.3 | 192.168.50.2     | 255.255.255.0 | 254      | 0.0.0.0     | 0.0.0.0     |
| ADMIN      | 192.168.40.1    | 192.168.60.3 | 192.168.40.2     | 255.255.255.0 | 254      | 0.0.0.0     | 0.0.0.0     |
| EN         | 192.168.30.1    | 192.168.60.3 | 192.168.30.2     | 255.255.255.0 | 254      | 0.0.0.0     | 0.0.0.0     |
| EE         | 192.168.20.1    | 192.168.60.3 | 192.168.20.2     | 255.255.255.0 | 254      | 0.0.0.0     | 0.0.0.0     |
| CS         | 192.168.10.1    | 192.168.60.3 | 192.168.10.2     | 255.255.255.0 | 254      | 0.0.0.0     | 0.0.0.0     |
| serverPool | 0.0.0.0         | 0.0.0.0      | 192.168.60.0     | 255.255.255.0 | 512      | 0.0.0.0     | 0.0.0.0     |

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix.:
    Physical Address.: 0001.6380.71D2
    Link-local IPv6 Address.: FE80::201:63FF:FE80:71D2
    IPv6 Address.: ::
    IPv4 Address.: 192.168.10.2
    Subnet Mask.: 255.255.255.0
    Default Gateway.: ::
                  192.168.10.1
    DHCP Servers.: 192.168.60.2
    DHCPv6 IAID.:
    DHCPv6 Client DUID.: 00-01-00-01-95-92-9D-B8-00-01-63-80-71-D2
    DNS Servers.: ::
                  192.168.60.3

Bluetooth Connection:

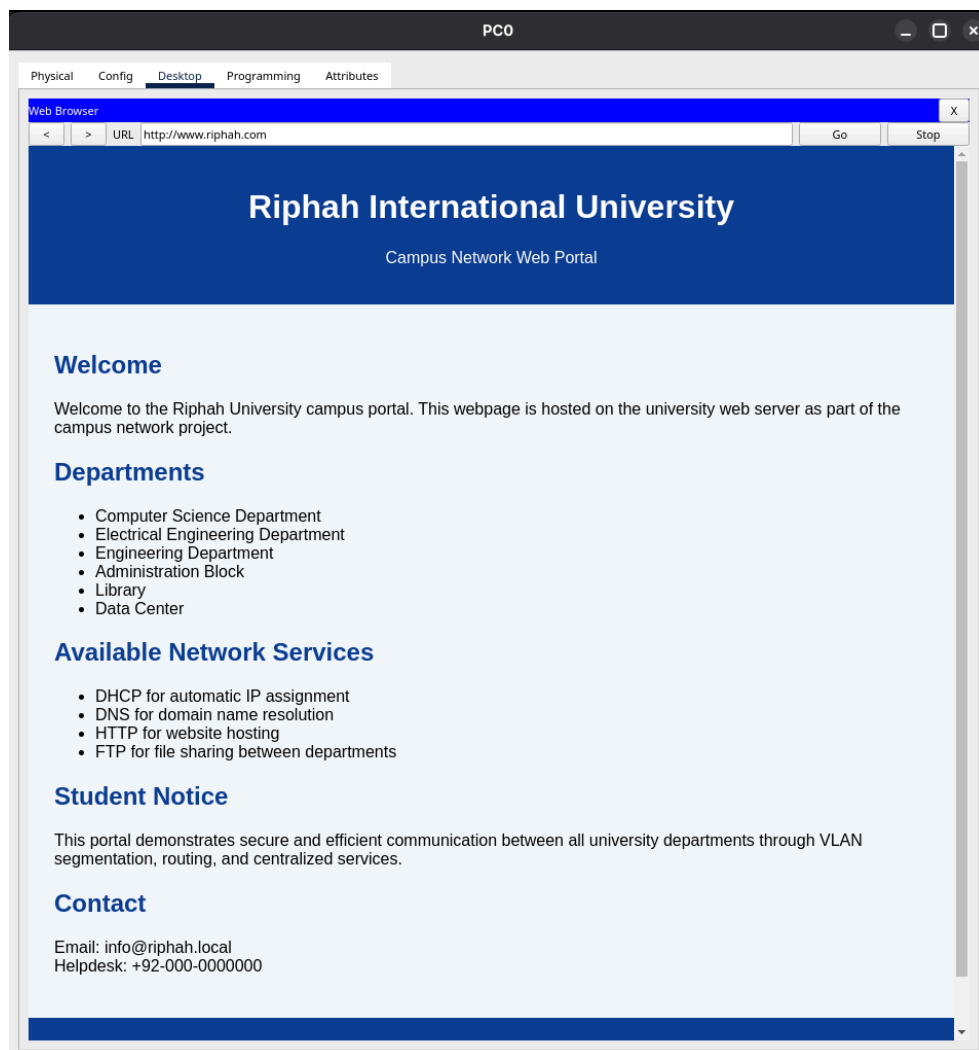
    Connection-specific DNS Suffix.:
    Physical Address.: 0030.A355.63E4
    Link-local IPv6 Address.: ::
    IPv6 Address.: ::
    IPv4 Address.: 0.0.0.0
    Subnet Mask.: 0.0.0.0
    Default Gateway.: ::
                  0.0.0.0
    DHCP Servers.: 0.0.0.0
    DHCPv6 IAID.:
    DHCPv6 Client DUID.: 00-01-00-01-95-92-9D-B8-00-01-63-80-71-D2
    DNS Servers.: ::
                  192.168.60.3
```

## 10. DNS and Web Server Configuration

The DNS server was configured to resolve local domain names for the university web and FTP services. Custom A records were created so that users could access services using domain names instead of IP addresses.

The web server hosted a simple Riphah University campus portal page. Clients successfully accessed the website in a browser using the configured DNS record, demonstrating correct DNS and HTTP operation.

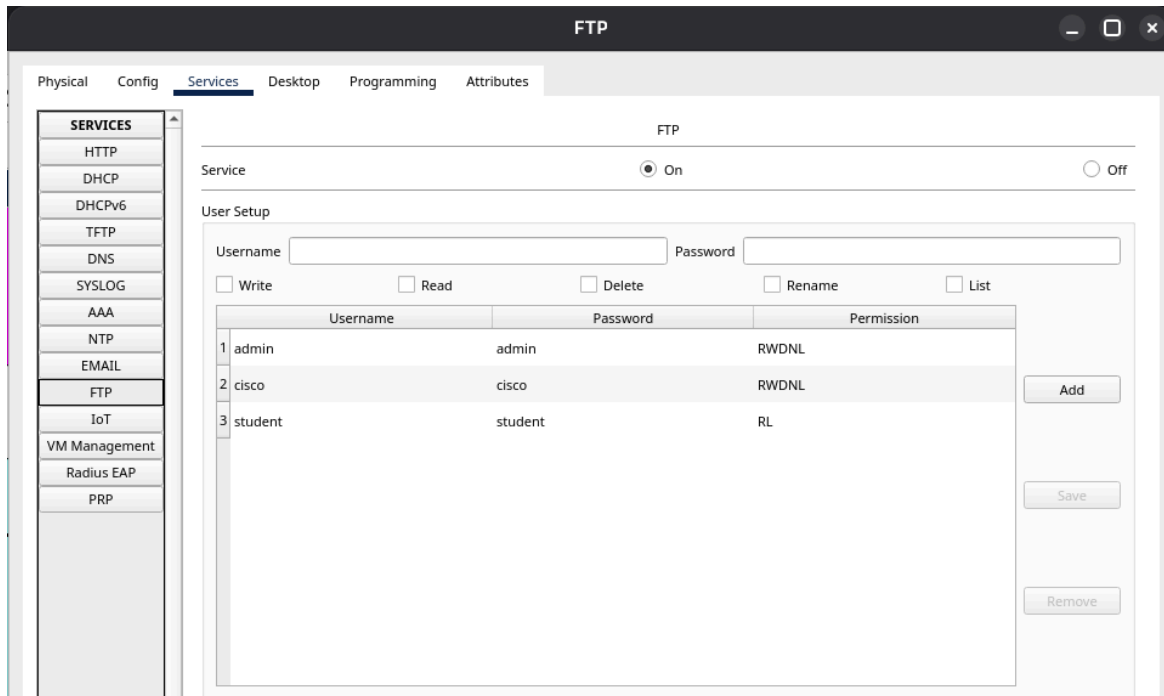
| No. | Name           | Type     | Detail       |
|-----|----------------|----------|--------------|
| 0   | ftp.riphah.com | A Record | 192.168.60.5 |
| 1   | www.riphah.com | A Record | 192.168.60.4 |



## 11. FTP Configuration

The FTP server was configured in the Data Center VLAN with user authentication enabled. Clients from other departments were able to connect to the server using both IP address and domain name, demonstrating functional file transfer service over the campus network.

Successful FTP login, directory listing, and upload/download operations verified that the FTP service was accessible across VLAN boundaries.



### FTP Server Configuration

```
C:\>ftp ftp.riphah.com
Trying to connect...ftp.riphah.com
Connected to ftp.riphah.com
220- Welcome to PT Ftp server
Username:student
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>get admin_card.txt

Reading file admin_card.txt from ftp.riphah.com:
File transfer in progress...

[Transfer complete - 15 bytes]

15 bytes copied in 0 secs
ftp>quit
```

### File downloading from FTP Server

## 12. ACL Configuration

Access Control Lists (ACLs) were used to restrict student-side VLANs from accessing the Administration VLAN. The ACLs were applied on the student VLAN interfaces to block unauthorized traffic close to the source while still allowing access to shared services such as DNS, Web, and FTP in the Data Center.

Testing confirmed that traffic to the Administration VLAN was blocked, while permitted traffic to the service servers continued to work normally.

```
C:\>ping 192.168.40.2

Pinging 192.168.40.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.40.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Ping to VLAN 40 is blocked

```
C:\>ping 192.168.50.3

Pinging 192.168.50.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.50.3: bytes=32 time<1ms TTL=125
Reply from 192.168.50.3: bytes=32 time<1ms TTL=125
Reply from 192.168.50.3: bytes=32 time=9ms TTL=125

Ping statistics for 192.168.50.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 3ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping to other VLAN's is working fine

## 13. Port Security

Port security was configured on access-layer switch ports connected to end devices. Sticky MAC addressing was used so that each secured port could automatically learn and retain the MAC address of the authorized host.

This configuration helps prevent unauthorized devices from being connected to the network through unused or reassigned access ports.

```
Switch(config)#interface range fa0/2 - 3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport port-security
Switch(config-if-range)#switchport port-security maximum 1
Switch(config-if-range)#switchport port-security violation restrict
Switch(config-if-range)#switchport port-security mac-address sticky
Switch(config-if-range)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
Switch#write memory
Building configuration...
[OK]
Switch#
```

### Port Security Configuration

```
Switch#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
          (Count)          (Count)          (Count)
-----
      Fa0/2          1          1          0      Restrict
      Fa0/3          1          1          0      Restrict
-----
Switch#show port-security address
      Secure Mac Address Table
-----
Vlan    Mac Address      Type                Ports    Remaining Age
      (mins)
-----
   10    0001.6380.71D2    SecureSticky        Fa0/2    -
   10    000C.CF97.60A1    SecureSticky        Fa0/3    -
-----
Total Addresses in System (excluding one mac per port)    : 0
Max Addresses limit in System (excluding one mac per port) : 1024
Switch#
```

## 14. Testing and Results

The network was tested in multiple stages to verify end-to-end functionality. Connectivity, automatic IP addressing, service access, access restrictions, and switch security were all validated successfully.

- Ping within the same VLAN was successful.
- Ping between permitted VLANs was successful.
- Ping from student VLANs to the Administration VLAN was blocked by ACLs.
- DHCP assigned correct IP settings to clients.
- DNS resolved local domain names correctly.
- The web page opened successfully in the client browser.
- FTP login and transfer worked across departments.
- Port security learned sticky MAC addresses on access ports.

| Fire                                                                                | Last Status | Source | Destination | Type | Color                                                                               | Time(sec) | Periodic | Num | Edit   | Delete   |
|-------------------------------------------------------------------------------------|-------------|--------|-------------|------|-------------------------------------------------------------------------------------|-----------|----------|-----|--------|----------|
|  | Failed      | PC0    | PC6         | ICMP |  | 0.000     | N        | 0   | (edit) | (delete) |
|  | Failed      | PC0    | PC6         | ICMP |  | 0.000     | N        | 1   | (edit) | (delete) |
|  | Successful  | PC0    | PC9         | ICMP |  | 0.000     | N        | 2   | (edit) | (delete) |
|  | Successful  | PC1    | PC3         | ICMP |  | 0.000     | N        | 3   | (edit) | (delete) |
|  | Successful  | PC4    | Web         | ICMP |  | 0.000     | N        | 4   | (edit) | (delete) |

Only transfers failed are from Student Department to Admin Department

## 15. Challenges Faced

Several practical issues were encountered during implementation, including VLAN trunk verification, inter-switch and router path configuration, DHCP relay setup, DNS refresh behavior in Packet Tracer, and FTP file handling. These issues were resolved through structured troubleshooting and staged testing.

The project demonstrated the importance of building a network in layers: physical topology first, then VLANs, routing, services, security, and finally testing.

## 16. Conclusion

This project successfully implemented a functional university campus network in Cisco Packet Tracer. The design combined VLAN-based segmentation, inter-VLAN routing, centralized services, and security controls to produce a realistic and well-structured campus environment.

The final network fulfilled the major technical requirements of the assignment, including DHCP, DNS, HTTP, FTP, ACL-based restriction, and port security. The implementation also provided hands-on experience with troubleshooting, design planning, and service integration in a simulated network environment.

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The source .pkt file can be found [here](#).

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